Karamveer Singh Sidhu

Deep Learning Researcher | Full-Stack Software Engineer

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Skills

Languages: TypeScript, JavaScript, Python, C++, Rust, SQL
Frontend/Backend: React.js, Next.js, Node.js, NestJS, REST APIs, BullMQ, OAuth, Web3, Design Systems
ML/Data/Infra: PyTorch, Flower, Federated Learning, R, Jupyter, PostgreSQL, MongoDB, Redis, Docker, AWS

Experience

Tokenbooks — June 2025 – Present
Software Engineer — *Vancouver, BC*

- Build full-stack Web3 finance features across wallet/exchange sync, accounting workflows, reporting, and payments.
- Shipped payment request intake, approval routing, counterparty/payment details, and crypto-to-fiat payout flows.
- Improved multi-wallet, multi-chain ingestion reliability through async job orchestration, status tracking, structured error handling, Coinbase OAuth reconnect flows, and Excel/XLSX exports.

University of Northern British Columbia — Sept. 2024 – June 2025
Graduate Teaching Assistant — *Prince George, BC*

- Designed and delivered labs for 110+ students covering statistics and visualization with R and R Commander.
- Developed Jupyter/Python labs, taught statistical modeling concepts, provided student support, and assisted with grading and exam invigilation.

CloudSEK — July 2022 – Aug. 2024
Software Development Engineer — *Bangalore, India*

- Built data visualization components, dashboards, and user-facing pages for XVigil and SVigil; contributed to in-house design system components.
- Implemented chunked APK uploads, reducing upload time by 40–50% while bypassing file size limits.
- Developed Exposure Search with other engineers, contributing to 100k+ searches and 3000+ leads; independently shipped a new product frontend within two weeks, supporting a $100k deal.

Research & Projects

MSc Thesis Research: Causal FL-IDS Defense — GitHub
Federated learning robustness for intrusion detection systems

- Designed a causal-defense CNN with a learnable sparse 64x64 causal matrix, aggregated across clients with FedAvg.
- Built a reproducible UM-NIDS pipeline and evaluated a 4-client non-IID PyTorch/Flower setup under FGSM and PGD attacks.
- Improved PGD robustness from 41.0% to 89.85% at epsilon = 0.20 with under 1% clean-accuracy loss, without adversarial training.

Education

University of Northern British Columbia — Sept. 2024 – June 2026
MSc, Computer Science — GPA: 4.08

Baba Banda Singh Bahadur Engineering College — 2018 – 2022
BTech, Computer Science — CGPA: 8.86

Publications

- **SoK: Privacy-aware LLM in Healthcare: Threat Model, Privacy Techniques, Challenges and Recommendations**. arXiv preprint, 2026.
- **Data science and analytics in agricultural development**. Environment Conservation Journal, 2021.
- **Advancements in farming and related activities with the help of artificial intelligence: a review**. Environment Conservation Journal, 2021.